/ Department of
Electrical Engineering

Control Systems Group

Systems and Control Integration Project 5SC26

Prof. dr. Siep Weiland

Version: April 16, 2026

Contents

1 Introduction

This student guide explains the goals and all aspects around the organization, requirements and assessment of the "Systems and Control Integration Project" with course code 5SC26. This is a compulsory course in the fourth quartile of the first year of the Masters program Systems and Control. This guide addresses the course as it is held in the academic year 2026-2027.

The course and project work starts with a kick-off meeting on Tuesday, April 21 and is finalized with the final exams on July 2 and July 3, 2026.

2 Course setup

2.1 General information

Course title:	Systems and Control Integration Project
Course code:	5SC26
Period:	Quartile 4 (21-04-2026 to 03-07-2026)
Degree program:	Graduate School TU/e, Masters program Systems and Control, Year 1, compulsory course in core of Systems and Control curriculum. The course cannot be taken as an elective course for other programs. The course is not recommended for Bachelor or Premaster students that are allowed to enter master courses in the fourth quartile, nor for students that enter the Systems and Control Masters program in February.
Type of education:	Full time. Project work carried out in teams throughout the quartile. This consists of experimental work during lab-sessions (to be scheduled by the teams).
Lecture hours:	One kick-off meeting on Tuesday, April 21, (13:30 - 15:15, Atlas -01.822). Instruction sessions and workshops on professional skills for (i) Academic writing, (ii) Presentation skills and (iii) Team organization and collaboration.
ECTS credits:	5
Course structure:	The Systems and Control Integration Project is a challenge-based learning course in which students collaborate in teams to integrate their knowledge on systems and control for the modelling, validation, control and performance analysis of a laboratory set-up. The control design and implementation has to be carried out in teams and should result in a presentation, a demonstration, a report and a defence of the work that is scheduled by the end of the quartile. Subject to departmental regulations, laboratory spaces are available after reservation to perform lab experiments. Some of the laboratory set-ups have a 24/7 remote access via the Remote Labs system. Technical support for all the laboratory set-ups is facilitated throughout the project. During the course, various trainings and workshops are held to improve or train professional skills on (i) academic writing, (ii) presenting and (iii) team organization and collaboration.
Lecturers:	prof. dr. Siep Weiland (responsible lecturer) dr. Leyla Ozkan prof. dr. Roland Toth ing. Will Hendrix (lab manager and lab support) ir. Peter den Toom (lab manager and technical support) Ir. Jake Rap (lab manager and technical support)

Technical help desk:	Ir. Peter den Toom (Lab access, technical support ball & plate, remote labs) Ing. Will Hendrix (lab access, technical support remote labs) Ir. Jake Rap (lab access, technical support crazy fly drones)
Course registration:	open November 13, 2025 up to and including March 22, 2026 un-enrolment until June 7, 2026.
Course material:	Slides kick-off meeting Instruction manuals and documentation for all lab set-ups. Instruction manuals and documentation on RemoteLabs Slides and readers on professional skills. All material is made available via Canvas
Responsible group:	Control Systems (CS) Department of Electrical Engineering
Secretariat:	Email: `secretariat.cs@tue.nl` Flux 5.132 tel.: 040 247 2300.

2.2 Brief summary

The Systems and Control Integration Project is a challenge based learning project in which students collaborate in teams to apply their knowledge on systems and control for the modeling, validation, control and analysis of a laboratory set up. A complete control design, its implementation and analysis has to be carried out by teams and should result in a presentation, a demonstration, a report and a defence of the work by the end of the quartile. During the quartile and as part of the course, various trainings are held on professional skills including trainings on academic writing, on presenting technical work and on the organization and collaboration of teamwork.

2.3 Learning objectives and content

The objective of the course is to have students collaborate in teams so as to apply and integrate material that is taught in the Systems and Control MSc curriculum so as to fully control a laboratory set-up. This is realized by combining the expertise of different team members (all students in the Systems and Control MSc programme) and by integrating the following learning objectives:

1. **Modeling**
   Objective is to apply one or more of the basic techniques to derive a mathematical model that represent the relevant dynamics of a specific laboratory set-up. Such a model can be derived from first principles and/or experimental data or a combination of both. Students can perform data acquisition, data processing, and can set up suitable experiments to identify physical parameters from experimental data. For experiment design, safety issues and system constraints are understood and taken into account. Students can make numerical implementations of mathematical models and are able to simulate model responses.

2. **Model validation**
   Objective is to simulate the acquired model and quantitatively assess its accuracy, validity and reliability on the basis of experimental data. Students show to be able to assess control relevant properties of the model including stability of fixed points, energy conservation, controllability, observability, stabilizability and/or detectability.

3. **Model-based control design**
   Objective is to formulate clear and unambiguous control objectives for the laboratory set-up. This formulation includes qualitative properties such as stability, safety of operation and robustness but also quantitative properties for optimised or ideal behaviour. Second objective on this aspect

involves the synthesis of a controller that achieves these goals and the demonstration that a synthesised controller actually achieves the performance objectives for the validated model.

4. **Implementation of controllers on a laboratory set-up**
   Objective is to implement the controller on the laboratory set-up in such a way that the quality of the controller and the control objectives can be assessed or tested on the basis of a suitably defined experiment on the laboratory set-up.

5. **Performance analysis and evaluation or verification of the design(s)**
   Objective is to carefully assess the properties and performance of the synthesised controller(s), by providing certifications on the performance and properties of the controlled system. These certifications include assessments on the stability, robustness, accuracy, operational constraints, limitations and quantified performance of the system. If multiple controllers are designed for the set-up, it is an objective to make a scientifically meaningful comparison between the performance of these controllers.

6. **Demonstration and reporting of the controlled set-up**
   Objective is to provide a demonstration of the controlled set-up showing the performance and performance limitations of the controlled system. In addition, all design choices can be explained and motivated in a presentation and in a detailed technical report that meets academic standards on clarity, structure, logic, objectivity, correctness and reproducibility.

It is an additional learning objective to train professional skills on academic writing, presenting technical material and the organization of teamwork. Specific learning objectives on these skills are the following:

1. **Academic writing**
   Students are able to report in written on a technical design that meets the international standards on readability, structure, reproducibility, grammatical and technical correctness, logical flow of reasoning, unambiguous and complete scientific referencing, conciseness and preciseness in expression and mathematical rigour.

2. **Presenting technical material**
   Students are able to deliver a presentation of technical work in front of an audience of peers and/or experts. The presentation is given at a proper pace, showing language proficiency, making use of available facilities, with adequate interaction with the audience, delivering the content in a structured and comprehensible manner.

3. **Organization and collaboration in a team**
   Students are able to organize work as a team, distribute tasks and plan research so as to execute and complete a team-task in a timely manner and meet deliverables. Students show commitment during team meetings, are able to share knowledge, are open with a positive attitude for constructive feedback to fellow students to enhance team performance, and are able to communicate with team members in an interdisciplinary manner to solve problems or to exchange ideas.

See Section 7 for more details on the professional skills.

2.4 Position in the curriculum, assumed previous knowledge

Position curriculum:	- Mandatory core course in the Systems and Control Masters curriculum.
Prior knowledge:	- Expertise from core, various specialization and elective courses covering different aspects on modelling, identification and control. - Programming experience in Matlab. - For some of the set-ups, programming experience in Python, ROS or C may prove useful.
Follow-up courses:	MSc courses in all fields of engineering, internship, graduation work.

With insufficient prior knowledge in the Systems and Control curriculum your team contribution is likely to be disproportionate to the work of other team members. In these cases we discourage you to register for or participate in this course.

2.5 Educational format

Educational format

The course has a *challenge-based format* which means that you as a student, in cooperation with your team, largely determines the direction, organization, planning and individual objectives to properly control a laboratory set-up of your choice. These set-ups do not have a pre-defined problem, objective or specific process leading to a well controlled configuration. The challenges associated with the set-ups are therefore largely open-ended where your ambitions can be focused on performance guarantees, robustness guarantees, operational ranges, user-friendliness, analytic certification of performance, or any combination of these properties.

As a team, you define your own (control) objectives and challenges and, as a team, you plan what it takes to work towards a solution. In doing so, different expertise, different perspectives and different experience from your team members can be used, explored, exploited and integrated to solve or tackle the challenges that you put forward as a team. An interdisciplinary approach is therefore the preferred route towards problem solving. This evidently requires commitment from all team members and a solid level of organization and collaboration within your team.

The course starts with a *kick-off meeting* that is held on Tuesday, April 21, (13:30 - 15:15, Atlas -01.822). There are 8 weeks planned to carry out the project in teams of maximally 4 students. Teams have regular (weekly or bi-weekly) progress meetings to discuss planning, organizational aspects, technical progress and division of tasks and responsibilities among the team members. Teams are responsible to define their own challenges in the project, to combine competences of team members, to organize lab-time for experimentation and testing, to realize a design process in cooperation among team members and to report on its outcome.

The project ends in the last week of the quartile with a presentation, a defence and a demonstration of the controlled laboratory set-up.

Professional skills

To facilitate the cooperation and to develop competences, professional skills on (i) Academic writing, (ii) Presentation of technical material and (iii) Team organization and collaboration are trained during various workshops that are held exclusively for students registered for this course. The workshops run in parallel with the project during the quartile. We refer to the schedule and calendar for details.

Attendance of the workshops is not mandatory, but highly recommended. Student attendance and/or team attendance at specific events will be monitored and registered. See Section 7 below.

Teams

Teams will be defined prior to the start of the course and consist of minimally 3, maximally 5 and ideally 4 students. For practical reasons, team compositions cannot be changed during the project. Teams are responsible for their own organization, planning and distribution of the work. Responsibilities and tasks among team members need to be clear, need to be documented and are evaluated. It is assumed that team members contribute in an equivalent manner to the project and collaborate on basically all aspects related to the project.

Some of the lab set-ups require dedicated skills on software and programming. For these set-ups we impose requirements on expertise of (some of the) team members.

Coaching and assistance

Technical questions on the use of RemoteLabs or technical questions on specific lab set-ups can be addressed to the *technical support staff*. Technical support is associated with the specific set-up you are working on. See Section 13 or Section 6 below for details. You can contact technical staff members by email or by appointment. Occasionally, the technical staff organizes separate and dedicated information sessions for teams. These coaching events, if any, will be announced via Canvas.

3 Deliverables, assessment and grading

Deliverables
Deliverables at the end of the fourth quartile are the following:

- **a report per team**
  The report consists of a quality documentation on the work performed in a maximum of *25 pages*. It describes the aspects of modeling, model validation, the control challenges that you defined for the set-up you are working on, the control design(s), their implementation, an analysis on the performance of the controlled laboratory set-up and a verification of the functioning of the design(s). A comparison of multiple designs is preferably included in the report. The report needs to meet the general scientific standards on proper technical writing, i.e., it needs to clearly, accurately, and concisely provide technical information on the work that has been carried out and provide proper references to other's work where appropriate. The report needs to have a cover page with the team members and their student ID's, and the report contains a section that indicates the individual contribution of team members to the project.

  The use and restrictions on AI tools in the writing of the report is described in Section 9 about Academic Integrity and AI usage.

- **a presentation per team**
  Every team organizes a 20 minutes presentation which covers the highlights of the designs. It is required that all team members present part of the work in a single presentation,

- **an oral defence of the work**
  A questioning on technical aspects that are documented in the report and the presentation will take place immediately after the presentation of the work.

- **a demonstration of the set-up**
  A demonstration is expected during or immediately after the defence of the work. The demonstration can be realized either on the lab set-up (in the laboratory) or as a video recording which clearly demonstrates the functioning of the system. Details on the way the demonstration needs to be performed will be communicated to you.

- **an evaluation of the team work**
  An assignment will be issued using the FeedbackFruits application in Canvas with dedicated questions on the organization and collaboration of your team.

Assessment and grading
The assessment of the course consists of the following seven parts:

1. Modeling and model validation (17.5% of the final grade)
2. Quality of control design (17.5% of the final grade)
3. Implementation of controller (17.5% of the final grade)
4. Performance analysis (17.5% of the final grade)
5. Quality of report and technical writing (10% of the final grade)

6. Quality of presentation (10% of the final grade)
7. Quality of organization and collaboration (10% of the final grade)

The weighted average will result in a team grade. Unless the contributions of team members are not equally distributed over the work, individual grades of team members will coincide with the team grade.

In addition to this *summative assessment*, also a number *formative assessments* take place during the quartile on the professional skills. Specifically,

- for the writing skills, a preliminary version of the report is handed in halfway the quartile and is commented upon for feedback on technical writing,
- a sample presentation is organized for feedback to team members and
- organizational and collaborative skills are discussed with coaches on the basis of a self-evaluation halfway the quartile.

These formative assessments take place on the professional skills prior to the summative assessment at the end of the quartile. See the schedule and calendar for details. The formative assessments do not lead to grades, and are not counting for the final assessment.

Your commitment to the course is important. The Program and Examination Regulations 2025-2026 (OER) of the Systems and Control Masters program state that this course has *no resits*. Failing the course therefore implies that you cannot re-do the course earlier than next year. See also Section 10 below.

4 Remote Labs

Some of the set-ups has remote access via the Remote-Labs environment. This means that the system is remotely accessible for experiments, data acquisition, real-time measurements and controller implementation though a web-based interface. Some important details on the use and functionality of the Remote-Labs system are summarized here.

What is Remote Labs?
The Remote Lab system is a web-based interface that allows you to interact with the physical set-up in the lab. The system allows you to login, and gives you real time access to experimental modes, allows you to tune variables, implement controllers, perform measurements and collect data resulting from the set-up. Remote-Labs consists of a web-based user interface for which software is delivered by and actively supported (also during this course) by the company Wolfpack in Eindhoven.

Scheduling and planning
Like in any experimental environment, lab-time is costly (or extremely costly) and need to be thoroughly planned for. With the use of remote labs this is not different. Scheduling of lab time is also necessary via the Remote Lab Scheduler, but does not depend on the opening times of the Department. We distinguish between two types of interaction with the Remote Lab system:

- **live experiments.** With a live experiment you directly operate and monitor the setup through manual interaction with the system. With a live experiment you run a real-time connection with the set-up over the user interface of the Remote-Lab manager. This means that you can perform experiments from your computer, acquire and log measured data and manually adjust parameters or implement controllers on the set-up during a live experiment. There is a camera on the system (and sufficient lighting if you work at night) to see and monitor the result of a live experiment on your computer. Data of your experiment will be logged by the Remote Lab system and is made available to you.

  Live experiments can be booked for periods of 30 minutes or 60 minutes. You will be disconnected from the system afterwards,

- **scheduled experiments.** A scheduled experiment is a fully automated and non-monitored set of tasks that the Remote Lab server performs on the set-up during a time slot that you schedule. This may be at night.

  The specifications of a scheduled experiment are uploaded to the Remote Lab server and performed and logged at the scheduled time. There exist a maximum time window that is available for a scheduled experiment.

Remote Labs and the set-up(s) connected to Remote Labs are available to you for 24 hours a day, and 7 days per week during the entire quartile. Down-times of the system because of maintenance will be announced in advance (assuming this is known to us). In principle, experiments and experiment time can be booked at any time and from any place, including nights and weekends.

Documentation

- Video on the use and functioning of Remote-Labs (posted on Canvas)
- Slides on Remote Labs. See the appropriate module on Canvas for further explanations on the Remote Lab.

Technical support

- Ir. Will Hendrix is the first responsible for all technical aspects on the implementation, functionality and functioning of Remote Labs.

5 Laboratory access

The lab setups are located in the

- **Control Systems Education (CSE) Lab**, location Flux 6.075 and Flux 6.075a
- **Autonomous Motion Control (AMC) Lab**, location Fenix 0.129.

During this quartile, it will be busy in the labs, not just with the Integration Project. As a consequence, lab-access is limited in time and only meant for doing measurements and experiments on the setups. The labs cannot be used for meetings or other activities. If you did not register for experimental work or are not involved in measurements you should not be there. If you are in one of the labs, you are allowed to work **only** on the process and set-up that is assigned to you. Refrain from touching, powering or operating any other equipment for your own safety. Moreover, you are not allowed to let other people enter the lab.

To get access to the labs, you need to submit an **Authorization Request to the Control Systems Labs** via this link. This authorization is fully digital , it addresses your awareness of the **Lab Safety Rules**, and provides access to the labs during this quartile by scanning a QR code on the main entrance of the lab.

For making reservations and booking time slots on the set-ups of the Ball and Plate system, the Gyroscope and for the Drones you need to make reservations through the *Clustermarket* syatem. For the Three Tank System, the *Remote Labs Scheduler* is available to book time slots. One student of each team will be granted access to Clustermarket. This will be initiated by Jake Rap and Peter den Toom as soon as student teams are known. As a team you can book 2 hour-slots of experiment-time and are limited to make reservations for one block at a time.

- Link to Clustermarket is Here. Login will be via SSO (single sign-on) via organisation *"Eindhoven University of Technology (TU/e)"*.

Lab managers of the Control Systems group are

- Jake Rap (Flux 5.135, Fenix), Autonomous Motion Control (AMC) Lab,
- Peter den Toom (Flux 5.135), Control Systems Education (CSE) Lab,
- Will Hendrix, (Flux 5.128), Remote Labs.

See Section 13 for their email addresses.

6 This year's lab set-ups

6.1 Three-Tank System

Background and purpose
Many fluid flow systems require liquid level control. Typical industrial applications of these control systems include waste water treatment, petro-chemical or oil-gas plants. The three tank system that we consider in this project is a rather typical case study representing such a fluid control system.

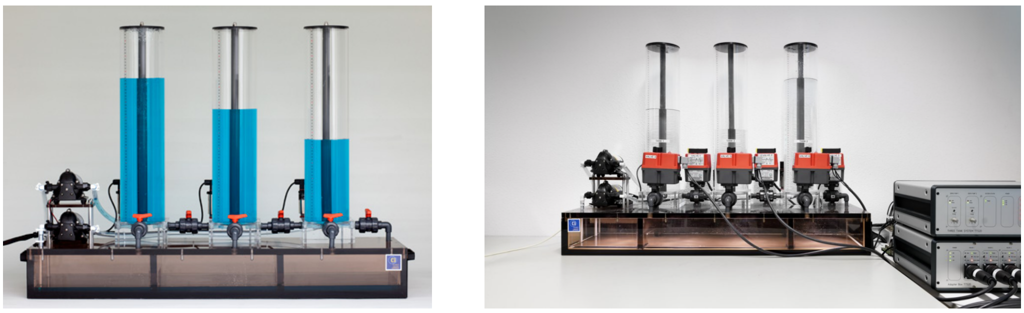

Figure 1: Three tank system

The system consists of three fluid flow tanks that are connected in series by cylindrical pipes at the bottom of the process. The middle tank has an outflow, while pumps control the inflow of the individual tanks. Purpose of the project is to track setpoints of the fluid hight in each of the tanks, subject to controlled and uncontrolled disturbances that represent leakages or liquid-flows at the bottom of the process. Key control questions involve the possibility of independent tracking of the liquid levels of one, two or three columns in the face of leakages and in the face of physical constraints that the system imposes. The system measures fluid levels in the three tanks and allows for actuation of pumps and valves to steer fluid levels.

Documentation

- Operator manual of the Three-Tank system with explanations on the process.
- RemoteLabs Introduction
- Dedicated software.

Access
The Three-tank system has remote access via Remote Labs and is also accessible on campus. The set-up is located in the Control Systems Education lab, room Flux 6.075. The lab is physically accessible on working days between 8:30 and 17:30 hrs. The Three-tank system has 24/7 remote access via Remote Labs.

Technical support
Dr. Leyla Ozkan is technical advisor for the Three-tank system. Technical support is given by Carlos José Gonzalez Rojas and Esteban Lage Cano.

6.2 3 DOF Gyroscope

Background and purpose

Gyroscopes are typically used for measuring or maintaining orientation of objects based on the principles of angular momentum. Applications of gyroscopes include inertial navigation systems mostly used in aircrafts, drones, rockets and ships, but also commercially available internal measurement units (IMUs) use miniature gyroscopes to measure velocity, acceleration, orientation and gravitational forces. Gyroscopes are also used as actuators for stabilization in ships or as devices to reject disturbances in high precision motion control applications.

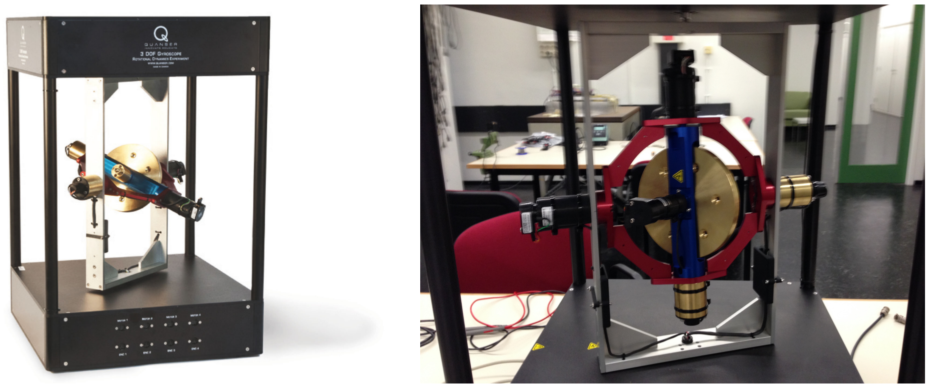

Figure 2: 3 DOF Gyroscope

This set-up involves developing a control solution for the position tracking of a 3 degrees-of-freedom (DOF) gyroscope with three concentric gimbals whose angular velocities can be controlled independently. The system allows to define very different performance specifications, and the purpose is to design a control architecture, to synthesize controllers and to verify the functioning of the system, its stability, and its tracking properties.

Documentation

- Gyroscope Operator Manual with technical specification, the dSPACe interface and the use of the control desk.
- Gyroscope Student Manual with the functioning of the control desk and the signal processing of the system.
- Gyro Dynamics Manual with help on the derivation of a mathematical model for the dynamics of a gyroscope with 3 degrees-of-freedom.
- A number of zip-files that help you with the simulation and visualization of the gyro dynamics.

Access

This set-up is in the Control Systems Education (CSE) laboratory in room Flux 6.075. See Section 5 for details. The lab is accessible on working days between 8:30 and 17:30 hrs.

Technical support

Dr. Roland Toth is responsible teacher for the teams working on the 3 DOF gyroscope. Technical support is given by ??.

6.3 Ball and Plate balancing system

Background and purpose

The Ball and Plate setup consists of three linear motors that are positioned in an equilateral triangle so as to control the angle and the height of a (horizontal) plate. Each of the linear motors has its own current-based controller that is an independent actuator with a separate power supply. Position measurements of the system take place via encoder information on the motor shafts. The three motors cause the plate to be lifted and tilted in independent directions, making the system basically a 6 DOF motion system. The aim of the system is to control the lifting and tilting of the plate so as to track a predefined position or a predefined trajectory of a free-moving ball on the plate. The position of the ball on the plate is measured via a camera system that is mounted above the set-up. The processing of the camera images may or may not be incorporated as challenge to improve the performance of the system.

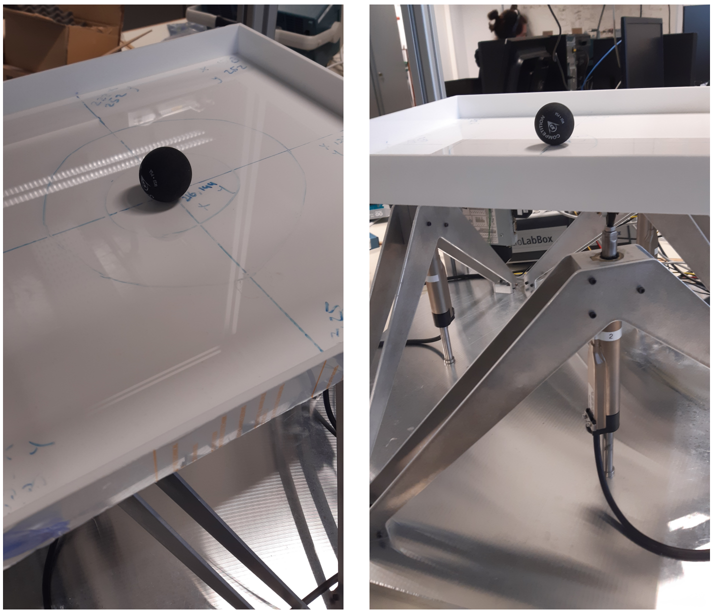

Figure 3: Ball and Plate Balancing System

Documentation

- Manual.
- Simulink Interface to the setup will be on the desktop of the setup.

Access
The Ball and Plate Balancing system is located in the Control Systems Education (CSE) laboratory, room Flux 6.075. For lab access, see Section 5. The lab is accessible at work days between 8:30 and 17:30 hrs.

Technical support
Technical support is given by Peter den Toom.

6.4 Crazy Flie drones

Background and purpose
Drones are Unmanned Aerial Vehicles (UAVs) that are piloted either remotely or are autonomous flying objects with onboard flight controllers. In principle, they are multipurpose flying platforms that can be utilized for a wide range of tasks from security surveillance, mapping, agricultural monitoring, building inspection, delivery, photography, filming, leisure and various military applications. With their relatively cheap construction costs and versatility of applications, research and development of drones is a rapidly emerging technological branch of the high-tech industry.

Prerequisites
Crazy-flie drone project preferred expertise in embedded programming (C), Python and ROS.

Figure 4: Crazy-Flie drones

Documentation

- A document on the functioning of the VICON sensor system.
- Different manuals on the drone hardware and software.
- The coaches of the drones will provide a tutorial session on the drones.

Access
This set-up is located in the *brand new* Autonomous Motion Control (AMC) laboratory of the Control Systems Group in the Fenix building. You will need to make lab-time reservations. The lab is accessible at work days between 8:30 and 17:30. See Section 5.

Technical support
Prof. dr. Siep Weiland is responsible teacher for the drones. Technical support is given by Jake Rap. Email address: `j.rap@tue.nl`

7 Professional skills

During this course you will receive active training and coaching for improving your professional skills on academic writing, on presenting techical work and on team organization and collaboration. This service is provided by the Education and Student Affairs (ESA) and is organized largely independent from the lecturers and coaches of the course.

Participation of the trainings of the professional skills is not mandatory. However, since this is the only place in the masters curriculum of the Systems and Control program where professional skills are trained, we highly recommend your participation in the workshops, and we do register your presence or absence in the workshops.

All professional skills are coordinated by Clarise Sky-Johnson. Details on the skills are the following.

7.1 Organization and collaboration within a team

Working in a team can be fun and rewarding but also requires skills that involve cooperation, organization, setting common goals, planning, and sharing responsibilities. Occasionally, differences of opinion, cultural differences or disagreements on what is generally understood by ‘common sense’ can be a cause of rather serious misunderstandings, mismatches of expectations or problems in the proper functioning of a team. The notion of "common sense" means different things to different people and may be much less "common" than what you always expected. To assist with team skills, ESA organizes a workshop on organizational aspects of team work, what it means and what it takes to collaborate, to share knowledge, to set common goals, to adhere to a planning, to distribute tasks and responsibilities and to deal with conflicts.

This workshop will be held per team and is scheduled on April 28 for a kick-off and on May 19 for a session on formative feedback. For the latter, a self-evaluation assignment need to be issued in the FeedbackFruits application on Canvas.

We expect you to participate in the workshop as a team. If you are unable to attend, you need to inform the responsible lecturer for this course.

7.2 Academic writing

Academic writing differs from non-academic writing in many aspects. The writing style for academic writing requires special focus on clarity, unambiguity, structure and conciseness, evidence, proof, reasoning, reproducibility of results, and exceptional clarity on conclusions. The skill of academic writing is of paramount interest in any technical report, documentation of technical results, scientific paper, thesis or dissertation, or scientific book. A proper training on the skill of academic writing will prove useful, not only for this course, but also for any future technical report that you need to write.

On *April 21 (16:30 - 17:30, Atlas -01.822)* a lecture will be given by Audrey Debije-Popson on specific aspects in academic writing. This to improve your skills on the writing of the report for this (and future) projects. The lecture is meant only for students registered for this course.

Attention is paid to the organization of the report, its structure, the relevant components, the expected contents of these components, concise and precise writing, readability, presentation of tables and figures, clarity of expression, formal grammar, the use of mathematics, mathematical rigor and logical correctness. In short, all relevant aspects for clear academic writing.

You will have the opportunity to receive feedback on a *preliminary version* of your report by uploading your draft report to the Canvas server at an early stage in the quartile. This should be done by *Friday, May 29, midnight*. It is recommended that all members of your team contribute to the writing of the (preliminary) report. The preliminary reports will be read and evaluated by the staff of ESA. A number of parallel sessions are organized on *June 2 and June 4* to provide individual feedback on your preliminary reports. The evaluation of the preliminary report is specifically meant as feedback and will not affect your grading for this course.

7.3 Presentation skills

A first general training on "Presenting like a Professional" will be given. As a preparation for your final presentations, you will receive a training on your presentation skills in the week prior to the final presentation and the demonstration of the set-up. The training focuses on do's and don'ts for technical presentations, how to make contact with your audience, how to prepare clear visual aids, and how to organize the content of your presentation. The trainings will be given as a workshop by coaches of ESA and will take place on *May 26 and June 25*. See the schedule for the precise details for your team. You will receive feedback on your presentation by staff members of ESA and through peer reviews. Time and location of the workshops are announced in the calendar below and on the course schedule on Canvas.

8 Team assignments

Team assignments and laboratory set-up assignments are listed below and represent the status of April 16, 2026.

Team number	Members	Lab set-up

9 Academic integrity and use of AI

Ground rules for students
Be aware that you should always operate with academic integrity, taking into account the rules and regulations about copyright, plagiarism and fraud. Use the links below to learn more about what you are (not) allowed to do in your assignments and assessments.

- See link on TU/e Rules for AI use in Education
- See link on TU/e regulations and Code of Conduct
- See link on TU/e Copyright Information Point
- See link on Program and Examination Regulations 2025-2026 (OER)

On the use of AI
Any work for an assignment in which generative AI tools are allowed, should make unambiguously clear:

- which tools you used,
- for which parts of the work you used them,
- what you used them for.

You should be critical about the quality of your information sources and make sure to precisely reference the parts of your work where AI tools were used.

> In this course you are not allowed to use generative AI to create content (like images, (source) code, video, text or any other kind). You are allowed to use AI-powered tools for improvements of your own content that operate at the level of at most a single sentence (e.g. spelling checkers) or a single line of code (as provided by many IDEs).

10 No resits

Due to the teamwork of the project, resits of (parts of) this course are very difficult to realize. It is for this reason that this course has no (formal) resits planned in the academic year (as also stated in the PER-2024-2025 of the Systems and Control Master's program). This means that the next opportunity to re-do the course is in the fourth quartile of the academic year 2025-2026. It is therefore important to adhere to the indicated deadlines of project deliverables. Failure to do so may result in one year of delay in the planning of your studies.

In the unfortunate event that you fail this course, we will investigate the reason for this and we may consider a repair trajectory. Such a trajectory consists of an additional assignment or a specific request to re-do part of the work. If possible and if necessary, these assignments will be given immediately after the final assessments.

11 Student wellbeing

TU/e aims to ensure that students are aware of what wellbeing means and feel empowered and enabled to take care of their wellbeing. Wellbeing is the responsibility of our community as a whole. Together we create a welcoming, safe and inclusive environment and provide the necessary knowledge and support to enable students to take control of their own wellbeing.

Full TU/e vision on wellbeing	Overview of all contact persons	THOR wellbeing support page

12 Scheduling and calendar

Important deadlines and events are summarized on the next page. The schedule is posted separately as a file on Canvas and under the 'Schedule' tag on Canvas. Last minute changes, if any, will be posted under the 'Schedule' tag on Canvas. The latest version of the schedule is therefore always posted under this tag.

Calendar

Systems and Control Integration Project (5SC26)

2026

Month	Date	Time	Location	Event	Meant for/status
April	21	13:30 - 15:15	Atlas -01.822	Kick-off event	Everybody
April	21	16:30 - 17:30	Atlas -01.822	Training Academic Writing (I)	Everybody
April	28	13:30 - 17:30	Atlas 07.201	Workshop Collaboration skills (I)	Teams 1,2,3,4
April	28	13:30 - 17:30	Atlas 08.201	Workshop Collaboration skills (I)	Teams 5,6,7,8
May	14	23:59:59	Canvas	**Submission 1st eval. Coll.&Org.**	Everybody
May	19	13:30 - 14:15	Gemini-Noord 1.859	Coaching session collab. skills (II)	Team 1
May	19	14:30 - 15:15	Gemini-Noord 1.859	Coaching session collab. skills (II)	Team 2
May	19	15:30 - 16:15	Gemini-Noord 1.859	Coaching session collab. skills (II)	Team 3
May	19	16:30 - 17:15	Gemini-Noord 1.859	Coaching session collab. skills (II)	Team 4
May	19	13:30 - 14:15	Gemini-Noord 1.860	Coaching session collab. skills (II)	Team 5
May	19	14:30 - 15:15	Gemini-Noord 1.860	Coaching session collab. skills (II)	Team 6
May	19	15:30 - 16:15	Gemini-Noord 1.860	Coaching session collab. skills (II)	Team 7
May	19	16:30 - 17:15	Gemini-Noord 1.860	Coaching session collab. skills (II)	Team 8
May	26	13:30 - 17:30	Metaforum Zaal 15	Training presentation skills (I)	All teams
May	29	23:59:59	Canvas	Deadline preliminary reports	All teams
June	2	13:30 - 14:15	on-line MS Teams	Feedback Academic writing	Team 1
June	2	14:30 - 15:15	on-line MS Teams	Feedback Academic writing	Team 2
June	2	15:30 - 16:15	on-line MS Teams	Feedback Academic writing	Team 3
June	2	16:30 - 17:15	on-line MS Teams	Feedback Academic writing	Team 4
June	4	08:45 - 09:30	on-line MS Teams	Feedback Academic writing	Team 5
June	4	09:45 - 10:30	on-line MS Teams	Feedback Academic writing	Team 6
June	4	10:45 - 11:30	on-line MS Teams	Feedback Academic writing	Team 7
June	4	11:45 - 12:30	on-line MS Teams	Feedback Academic writing	Team 8
June	25	08:45 - 10:30	Flux 7.177.	Training presentation skills (II)	2 groups of 4 teams
June	25	10:45 - 12:30	Flux 7.177	Training presentation skills (II)	2 groups of 4 teams
June	28	23:59:57'	Canvas	**Submission final report**	All teams
June	30	23:59:59	Canvas	**Submission 2nd eval. Collab.**	Everybody
July	2	08:30 - 17:00	Flux 0.150	Final assessments	per team
July	3	08:30 - 13:00	Flux 5.256	Final assessments	per team
July	3	10:30 - 17:00	Flux 0.150	Final assessments	per team

Table 1: Schedule of activities (version March 26, 2026).

13 Who is who?

Who	Address	Email	Function
Siep Weiland	Flux 5.134	secretariaat.cs@tue.nl	responsible lecturer
Roland Toth	Flux 5.134	r.toth@tue.nl	Gyroscope system
Leyla Ozkan	Flux 5.128	l.ozkan@tue.nl	Three-tank system
Peter den Toom	Flux 5.135	p.d.toom@tue.nl	Ball and Plate system
Jake Rap	Flux 5.135	j.e.w.rap@tue.nl	Drones technical support
Will Hendrix	Flux 5.128	w.h.a.hendrix@tue.nl	Remote labs, technical support
Carlos J. Gonzalez Rojas	Flux 5.130	c.j.gonzalez.rojas@tue.nl	Three-tank system technical support
Esteban Lage Cano	Flux 5.130	e.c.lage.cano@tue.nl	Three-tank system technical support
Clarise Sky-Johnson	Traverse	c.c.l.sky.johnson@tue.nl	Coordinator professional skills
Wouter Mommers	ESA	w.p.f.mommers@tue.nl	Coach presentation skills
Erik Roebroek	ESA	e.j.w.roebroek@tue.nl	Coach presentation skills
Audrey Debije-Popson	ESA	a.m.m.debije.popson@tue.nl	Coach academic writing
Anouk van der Kerkhof	ESA	a.c.p.v.d.kerkhof@tue.nl	Coordinator collaboration skills

14 Feedback and evaluations

Comments and feedback can be provided through the Canvas forum, by e-mail or in person. As control specialists, we appreciate both positive and negative feedback as both types of feedback can stabilize. Because of the rather special character of the course, we highly appreciate if you take the time to evaluate the course after the final gradings. In doing so, you contribute in improving the MSc curriculum Systems and Control for students after you.